



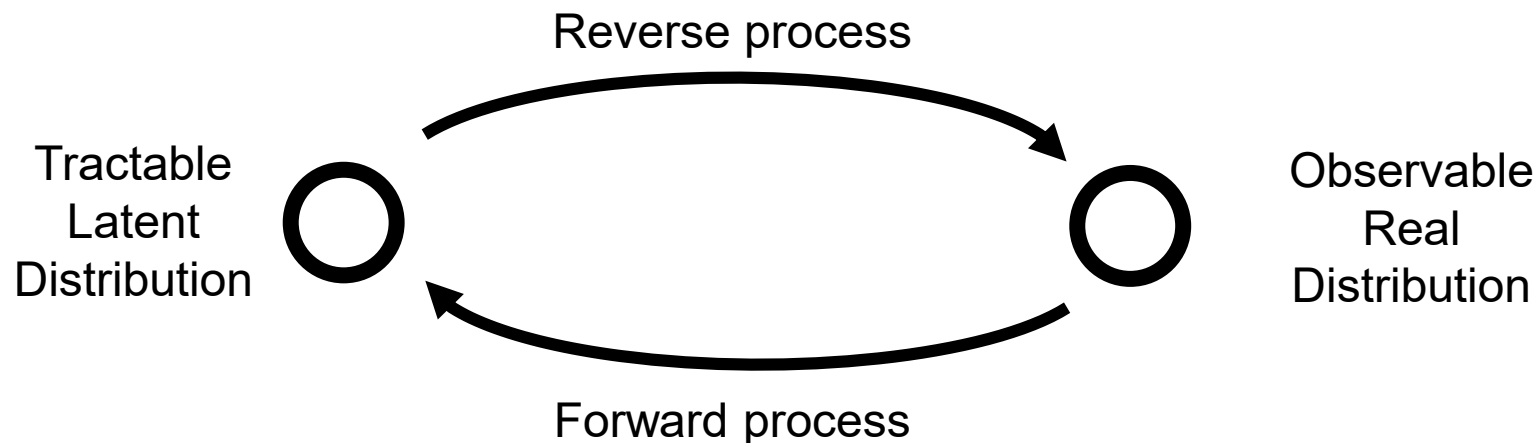
Diffusion Model: **Application, Method, and** **Implementation**

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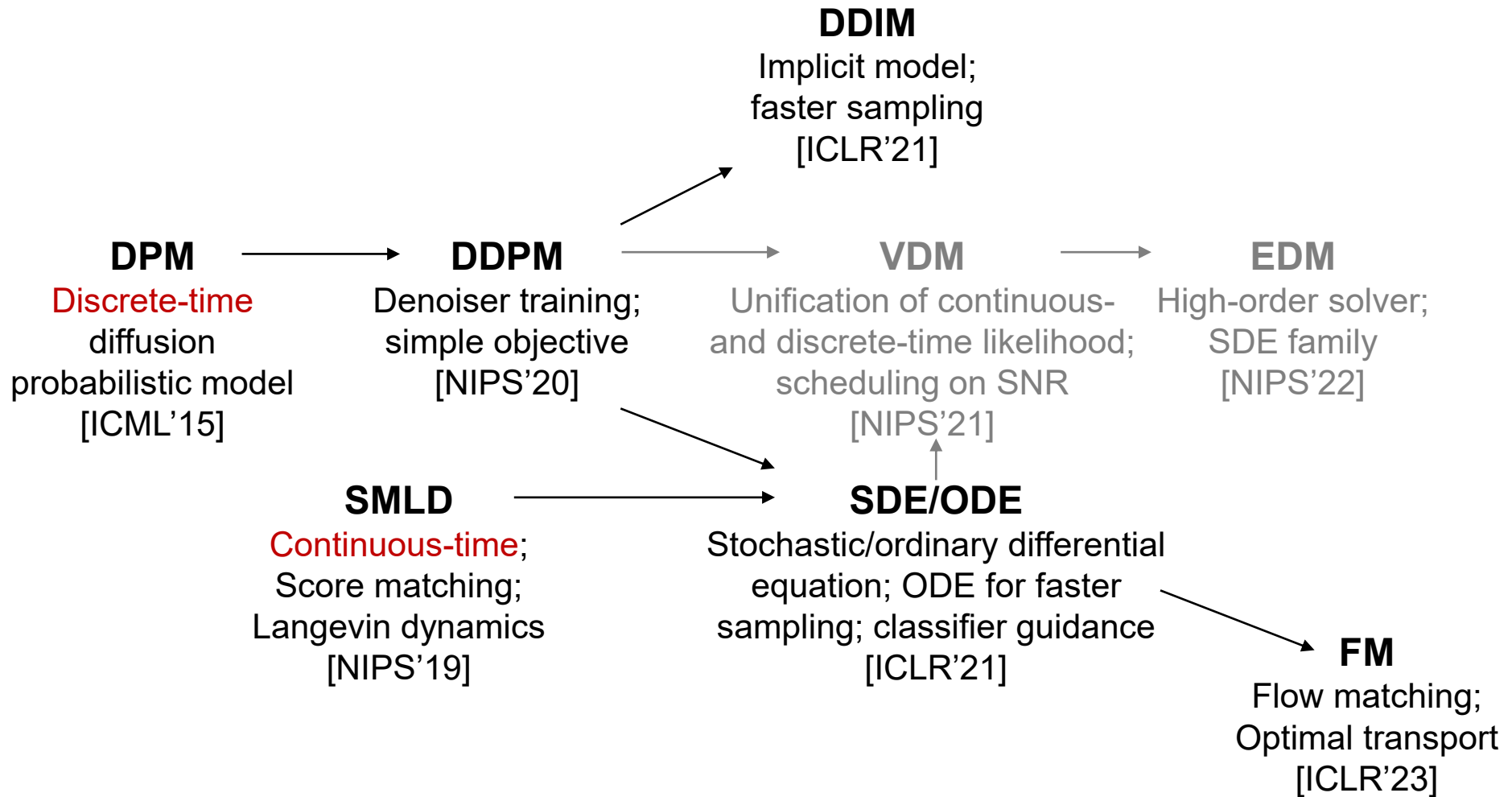
A diffusion model, in general, comprises three key components—forward process, reverse process, and sampling procedure (based on chosen framework)—with the following typical setting:



Method	Framework	Forward Process	Reverse Process
DDPM	Discrete-time Probabilistic Model	Stochastic, Markovian	Stochastic, Markovian
DDIM		Stochastic, Non-Markovian	Implicit, Markovian
SMLD	Continuous-Time Langevin Dynamics	Stochastic, Non-Markovian	Stochastic, Markovian
SDE	Continuous-Time Differential Equation	Stochastic, Markovian	
ODE		Deterministic, Invertible	
FM			



Outline





Outline

- **Method**
 - **DDPM/DDIM**
 - SMLD
 - SDE/ODE
 - FM
 - Appendices for equations



Diffusion Probabilistic Model (DPM, ICML'15)

(1/2)

- Matches data distribution $q(x_0)$ using generative distribution $p_\theta(x_0)$ under diffusion framework
 - Forward diffusion gradually adds noise (fixed Markov process):

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

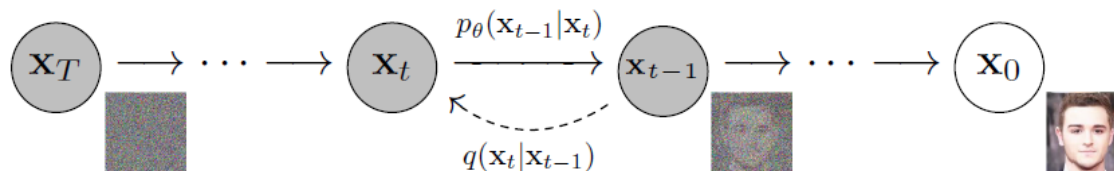
$$\Rightarrow q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad \alpha_t := 1 - \beta_t$$

Variance-preservative

$$\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$$

- Reverse generation gradually recovers intermediate distributions (stochastic Markov process):

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \underbrace{\boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{x}_t, t)}_{\text{Learn parameterized NNs}})$$



Credit: Song et al., NeurIPS'20; DDPM.



Diffusion Probabilistic Model (DPM, ICML'15)

(2/2)

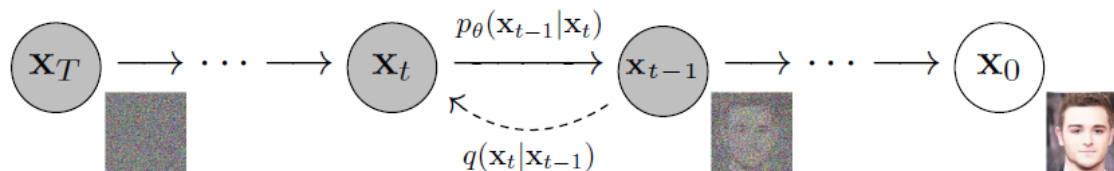
- Trains μ_θ and Σ_θ by optimizing an upper bound of intractable NLL
 - Evidence lower bound (ELBO) for x with auxiliary z :

$$\log p_\theta(x) = \log \int p_\theta(x, z) dz = \log \mathbb{E}_{q(z|x)} \left[\frac{p_\theta(x, z)}{q(z|x)} \right] \geq \mathbb{E}_{q(z|x)} \left[\log \frac{p_\theta(x, z)}{q(z|x)} \right]$$

- Variational upper bound for NLL with auxiliary $\mathbf{x}_{1:T}$:

$$\mathbb{E} [-\log p_\theta(\mathbf{x}_0)] \leq \mathbb{E}_q \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right] = \mathbb{E}_q \left[-\log p(\mathbf{x}_T) - \sum_{t \geq 1} \log \frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})} \right] =: L$$

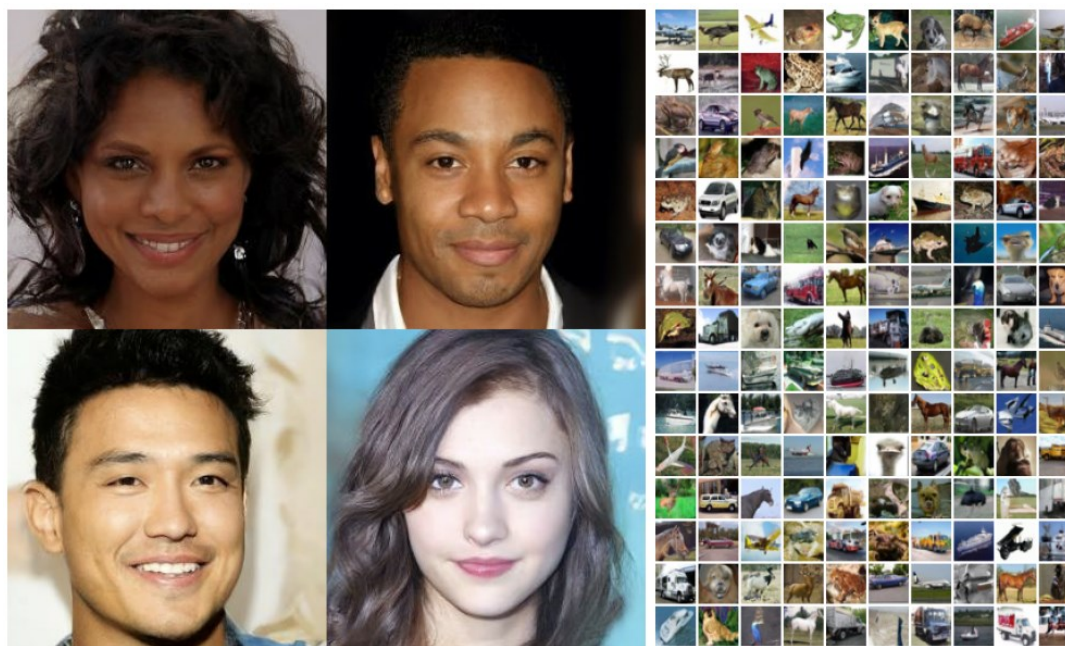
$\mu_\theta, \Sigma_\theta$: time-variant!



Credit: Song et al., NeurIPS'20; DDPM.

Denoising Diffusion Probabilistic Model (DDPM, NIPS'20)

- Starting point of the wave of diffusion models
- Contribution
 - Implement μ_θ via a single t-embedded denoising model ϵ_θ (with constant Σ_θ)
 - Propose a simple objective (equivalent to score matching)
 - Apply large U-Net, e.g. 36M for CIFAR-10, 114M for CelebA-HQ/LSUN





DDPM formulation (1/3)

- Better formulation for variational bound (from DPM)

$$\mathbb{E}_q \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))}_{L_T} + \sum_{t>1} \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} \underbrace{- \log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0} \right]$$

- Simple form of L_{t-1} for Gaussians

Major term: learn the information of x_0
by **time-embedded** μ_θ

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}), \quad \Rightarrow \quad L_{t-1} = \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \|\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2 \right] + C$$

where $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$

and $\tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$

Common choice for KLD
minimization when Σ_θ is not trainable.
Compatible to SDE.

$$\Sigma_\theta(\mathbf{x}_t, t) = \sigma_t^2 \mathbf{I}$$

$$\sigma_t^2 = \tilde{\beta}_t \quad \text{or} \quad \sigma_t^2 = \beta_t$$

Constant for fixed
pre-defined covariance
matrices

Optimized solution for the case in which the source entropy is the minimum (lower bound): $q(x_0) = \delta(x_0 - x')$ for one single image x'

Optimized solution for maximum source entropy (upper bound): $q(x_0) \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for AWGN



DDPM formulation (2/3)

- Denoising formulation through reparameterization
 - Uses time-embedded ϵ_θ to estimate the noise ϵ at each t
 - Shares parameters over t

$$\begin{aligned}
 L_{t-1} &= \mathbb{E}_q \left[\frac{1}{2\sigma_t^2} \left\| \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t) \right\|^2 \right] + C \\
 &\quad \text{reparameterize } \tilde{\mu}_t \text{ to eliminate } x_0 \quad \downarrow \quad \mathbf{x}_t(\mathbf{x}_0, \epsilon) = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \text{ for } \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \\
 L_{t-1} - C &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{1}{2\sigma_t^2} \left\| \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t(\mathbf{x}_0, \epsilon) - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) - \mu_\theta(\mathbf{x}_t(\mathbf{x}_0, \epsilon), t) \right\|^2 \right] \\
 &\quad \text{mimic } \tilde{\mu}_t \quad \downarrow \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right) \\
 &= \mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]
 \end{aligned}$$



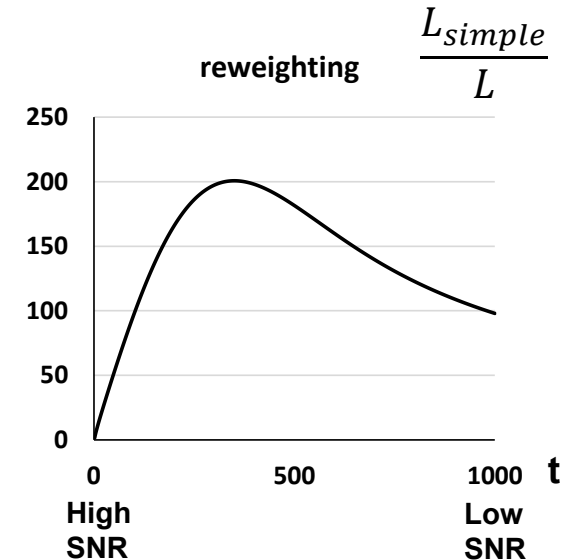
DDPM formulation (3/3)

- Simple objective

$$\mathbb{E}_{\mathbf{x}_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2 \alpha_t (1 - \bar{\alpha}_t)} \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

reweighting

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, \mathbf{x}_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$



- Same and independent training behavior over time
- Diffusion step number $T = 1000$ (both for training and inference)
- linearly scale β_t from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$
- Note: L_{simple} highlights more around mid-SNR, then low-SNR, and suppresses high-SNR => good for fidelity; not good for likelihood



Denoising Diffusion Implicit Model (DDIM, ICLR'21)

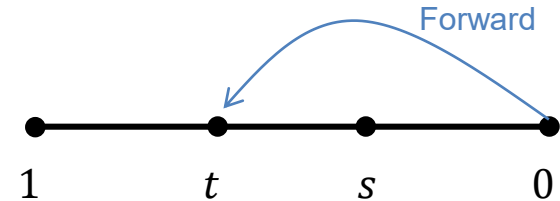
- Converts a pre-trained DDPM into implicit/deterministic generation and enables merely 20-100 steps based on
 - Observation
 - Forward diffusion is unnecessary: the Markovian forward process of $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ is not actually required for DDPM training and inference
 - Only $q(\mathbf{x}_t|\mathbf{x}_0)$ matters for the estimation of ϵ
 - Approach
 - 1) Reformulate a non-Markovian forward process while keeping $q(\mathbf{x}_t|\mathbf{x}_0)$ the same
 - 2) Construct a reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ to match the corresponding $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ and derive the same simple denoising objective (but with different reweighting)
 - 3) Make $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ implicit by setting the covariance matrices to zero
 - 4) Accelerate inference by sampling with a sub-sequence of $[1:T]$, e.g. $[1:40:T]$
 - Note: DDPM can sample with a sub-sequence as well but results in poorer quality



Unified formulation for DPM (1/2)

- Forward process (not necessarily Markovian)

- Pre-defined scheduling by (α_t, σ_t)
- General scheduling: SNR decreasing, i.e. $\frac{\alpha_t}{\sigma_t} < \frac{\alpha_s}{\sigma_s}$ for $t > s$



equivalent

$$q(\mathbf{x}_s | \mathbf{x}_0) = N(\alpha_s \mathbf{x}_0, \sigma_s^2 I)$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = N(\alpha_t \mathbf{x}_0, \sigma_t^2 I)$$

$$\mathbf{x}_s = \alpha_s \mathbf{x}_0 + \sigma_s \boldsymbol{\epsilon}_s, \boldsymbol{\epsilon}_s \sim N(0, I)$$

$$\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}_t, \boldsymbol{\epsilon}_t \sim N(0, I)$$

$$q(\mathbf{x}_s | \mathbf{x}_t, \mathbf{x}_0) = N\left(\alpha_s \mathbf{x}_0 + \sqrt{\sigma_s^2 - \eta_{s,t}^2} \frac{\mathbf{x}_t - \alpha_t \mathbf{x}_0}{\sigma_t}, \eta_{s,t}^2 I\right)$$

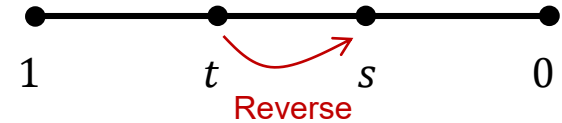
$$\mathbf{x}_s = \alpha_s \mathbf{x}_0 + \sqrt{\sigma_s^2 - \eta_{s,t}^2} \boldsymbol{\epsilon}_t + \eta_{s,t} \boldsymbol{\epsilon}_{s,t}, \boldsymbol{\epsilon}_{s,t} \sim N(0, I)$$

- Simple training objective: $L_{\text{simple}}(\theta) = \mathbf{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}_t} \left[\left\| \boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta^{(t)}(\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}_t) \right\|^2 \right]$
- Special cases:
 - DDIM (Non-Markovian): $\eta_{s,t}^2 = 0$
 - DDPM (Markovian): $\eta_{s,t}^2 = \sigma_s^2 \left(1 - \frac{\alpha_t^2 \sigma_s^2}{\sigma_t^2 \alpha_s^2}\right)$

Model trained for scheduling with (α_t, σ_t)



Unified formulation for DPM (2/2)



- Reverse process with ancestral sampling (Markovian)
 - Derive $\mathbf{x}_s$ by predicted data $\mathbf{x}_\theta^{(t)}$ or noise $\epsilon_\theta^{(t)}$ at the previous time step t
 - Prediction equivalence: $\mathbf{x}_t = \alpha_t \mathbf{x}_\theta^{(t)}(\mathbf{x}_t) + \sigma_t \epsilon_\theta^{(t)}(\mathbf{x}_t)$

$$p_\theta(\mathbf{x}_s | \mathbf{x}_t) = N(\alpha_s \mathbf{x}_\theta^{(t)} + \sqrt{\sigma_s^2 - \eta_{s,t}^2} \epsilon_\theta^{(t)}, \eta_{s,t}^2 I) \quad \longleftrightarrow \quad \boxed{\mathbf{x}_s = \alpha_s \mathbf{x}_\theta^{(t)} + \sqrt{\sigma_s^2 - \eta_{s,t}^2} \epsilon_\theta^{(t)} + \eta_{s,t} \epsilon_{s,t}, \epsilon_{s,t} \sim N(0, I)}$$

General update rule

Special cases:

- DDIM (implicit): $\eta_{s,t}^2 = 0 \Rightarrow \mathbf{x}_s = \alpha_s \mathbf{x}_\theta^{(t)} + \sigma_s \epsilon_\theta^{(t)} = \frac{\sigma_s}{\sigma_t} \mathbf{x}_t + \alpha_s (1 - e^{-h}) \mathbf{x}_\theta^{(t)} = \frac{\alpha_s}{\alpha_t} \mathbf{x}_t - \sigma_s e^h (1 - e^{-h}) \epsilon_\theta^{(t)}$ (data prediction)
- DDPM (stochastic): $\eta_{s,t}^2 = \sigma_s^2 (1 - e^{-2h}) \Rightarrow \mathbf{x}_s = \mu_\theta^{(t)} + \eta_{s,t} \epsilon_{s,t}$ where $\mu_\theta^{(t)} = \alpha_s \mathbf{x}_\theta^{(t)} + e^{-h} \sigma_s \epsilon_\theta^{(t)}$ (noise prediction)

$$\begin{aligned} \mathbf{x}_\theta^{(t)} &= \frac{\mathbf{x}_t - \sigma_t \epsilon_\theta^{(t)}}{\alpha_t} \\ \mathbf{s}_\theta^{(t)} &= -\frac{\epsilon_\theta^{(t)}}{\sigma_t} \end{aligned}$$

Prediction model
conversion

$$\begin{aligned} h &= \log \frac{\alpha_s}{\sigma_s} - \log \frac{\alpha_t}{\sigma_t} > 0 \\ e^{-h} &= \frac{\alpha_t}{\sigma_t} \frac{\sigma_s}{\alpha_s} < 1 \\ e^h &= \frac{\alpha_s}{\sigma_s} \frac{\sigma_t}{\alpha_t} > 1 \end{aligned}$$

h : log-SNR increment for reverse process

$$\text{(data prediction)} = \frac{\sigma_s}{\sigma_t} e^{-h} \mathbf{x}_t + \alpha_s (1 - e^{-2h}) \mathbf{x}_\theta^{(t)}$$

$$\text{(noise prediction)} = \frac{\alpha_s}{\alpha_t} \mathbf{x}_t - \sigma_s e^h (1 - e^{-2h}) \epsilon_\theta^{(t)}$$

$$\text{(score prediction)} = \frac{\alpha_s}{\alpha_t} \mathbf{x}_t - \frac{\alpha_s}{\alpha_t} (1 - e^{-2h}) \mathbf{s}_\theta^{(t)}$$



CIFAR10 and CelebA image generation measured in FID

$$\eta_{S,t}^2 = \eta^2 \cdot \sigma_S^2 (1 - e^{-2h})$$

S	CIFAR10 (32×32)					CelebA (64×64)						
	10	20	50	100	1000	10	20	50	100	1000		
η	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51	DDIM
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64	
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28	
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98	
$\hat{\sigma}$	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26	$\sigma_t^2 = \tilde{\beta}_t$ $\sigma_t^2 = \beta_t$	

- DDIM provides good quality down to 10 steps
- DDPM struggles even with 100 steps but achieves the best quality with 1000 steps and higher injected noise $\sigma_t^2 = \beta_t$ (note: this configuration is not compatible to the general update formulation)
- Issues for DPM formulation
 - Unable to explain why DDIM works even when the NLL bound explodes as $\sigma_t^2 = 0$
 - Unable to explain why DDIM is good at small steps while DDPM fails



Summary for Discrete-time DPM

- One forward process with specific scheduling of $\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \epsilon_t$ corresponds to numeric reverse processes of different $\eta_{s,t}$
 - Unified formulation for both stochastic and deterministic reverse processes
- Different prediction networks are interchangeable
 - Noise-/data-/score-prediction (and v-prediction)
 - Each one presents different reweighting for training loss
- Many training hyperparameters need to be set up
 - Scheduling of α_t, σ_t ; reweighting w_t ; prediction type
- Implicit generation outperforms for few steps based on empirical observation without analytical reasoning



Outline

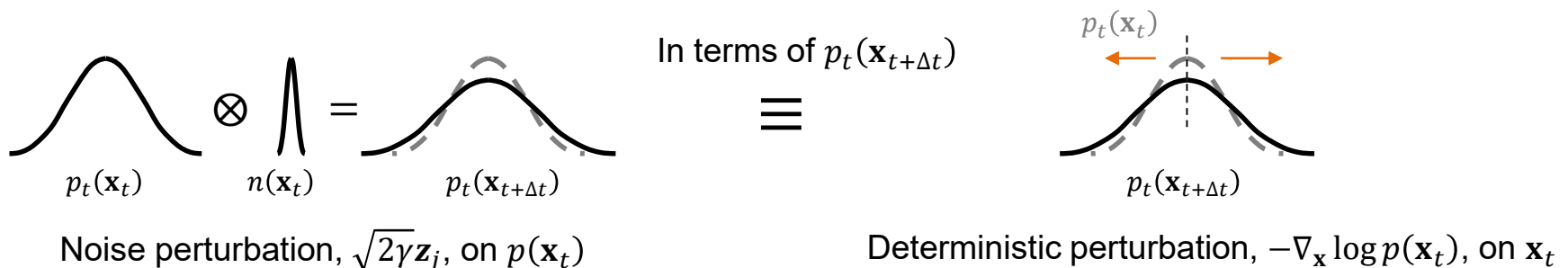
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Score matching with Langevin dynamics (SMLD, NIPS'19) (1/2)

- Langevin dynamics
 - More precisely, Langevin Monte Carlo (LMC) sampling
 - Useful for energy-based model (directly learns $\log p_\theta(\mathbf{x})$) to generate samples through a stochastic process
 - Monte Carlo simulation of the following equation converges to time-invariant distribution $q(\mathbf{x})$ with **score function** $\nabla_{\mathbf{x}} \log q(\mathbf{x})$
 - i.e. $p(\mathbf{x}_j) \approx p(\mathbf{x}_{j-1})$ as $\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}_{j-1}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x}_{j-1}) \Rightarrow p(\mathbf{x}_j) \approx p(\mathbf{x}_{j-1}) \approx p_\theta(\mathbf{x}_{j-1})$

$$\mathbf{x}_j \leftarrow \mathbf{x}_{j-1} + \gamma \nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}_{j-1}) + \sqrt{2\gamma} \mathbf{z}_j \text{ where } \mathbf{z}_j \sim \mathcal{N}(0, \mathbf{I})$$





Score matching with Langevin dynamics (SMLD, NIPS'19) (2/2)

- Noise conditional score network
 - Considers a decreasing sequence of noise scales $\{\sigma_i\}$ and the corresponding perturbed data distributions $q_{\sigma_i}(\mathbf{x}_i|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_i^2 \mathbf{I})$
 - Trains a score network $\mathbf{s}_\theta(\mathbf{x}, \sigma)$ for score matching at each σ_i :

$$L = \mathbf{E}_{\mathbf{x}_i, \sigma_i, \mathbf{x}_0} \left[\sigma_i^2 \left\| \mathbf{s}_\theta(\mathbf{x}_i, \sigma_i) - \nabla_{\mathbf{x}_i} \log q_{\sigma_i}(\mathbf{x}_i|\mathbf{x}_0) \right\|^2 \right]$$
$$\sigma_i^2 = 1 / \mathbf{E}_{\mathbf{x}_i, \sigma_i} \left[\left\| \nabla_{\mathbf{x}_i} \log q_{\sigma_i}(\mathbf{x}_i|\mathbf{x}_0) \right\|^2 \right]$$

- The gradient of the above loss, formed by aggregation of conditional score loss, equals to the gradient of the intractable score loss $\mathbf{E}_{\mathbf{x}_i, \sigma_i} \left[\sigma_i^2 \left\| \mathbf{s}_\theta(\mathbf{x}_i, \sigma_i) - \nabla_{\mathbf{x}_i} \log q_{\sigma_i}(\mathbf{x}_i) \right\|^2 \right]$
 - Generates samples sequentially with $\sigma_{i=1:T}$ and for each σ_i performs LMC sampling with estimated score $\mathbf{s}_\theta(\mathbf{x}_i, \sigma_i)$
- Comparison to DDPM
 - Same denoising loss $L = \mathbf{E}_{\mathbf{x}_i, \sigma_i, \mathbf{x}_0} [\left\| \epsilon_\theta(\mathbf{x}_i, \sigma_i) - \epsilon_i \right\|^2]$ if $\epsilon_\theta = -\sigma_i \mathbf{s}_\theta$, $\epsilon_i = \mathbf{x}_i - \mathbf{x}_0$
 - Different generation process (multi-step correction for each σ_i)
 - Different forward process since $\mathbf{x}_i$ is variance-explosive



Implication from SMLD

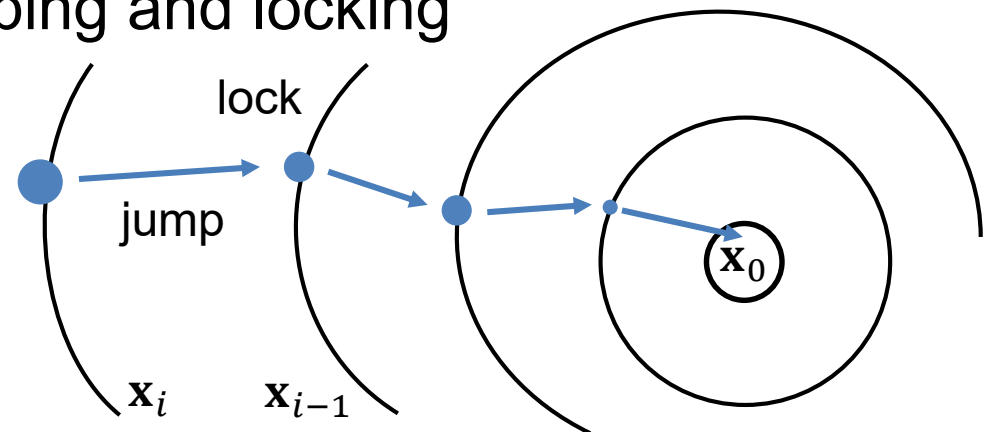
- Best score predictor learns the expected value of conditional score estimation that equals to the target score

$$\mathbf{s}_{\theta}^{best}(\mathbf{x}, \sigma_i) = \int q_{\sigma_i}(\mathbf{x}_0|\mathbf{x}) \nabla_{\mathbf{x}} \log q_{\sigma_i}(\mathbf{x}|\mathbf{x}_0) d\mathbf{x}_0 = \nabla_{\mathbf{x}} \log q_{\sigma_i}(\mathbf{x})$$

- Therefore, best noise predictor learns the expected value of the conditional noise $\epsilon = \mathbf{x} - \mathbf{x}_0$ given a latent $\mathbf{x}$

$$\epsilon_{\theta}^{best}(\mathbf{x}, \sigma_i) = \int q_{\sigma_i}(\mathbf{x}_0|\mathbf{x}) \cdot \epsilon(\mathbf{x}, \mathbf{x}_0) d\mathbf{x}_0$$

- Updates like manifold jumping and locking



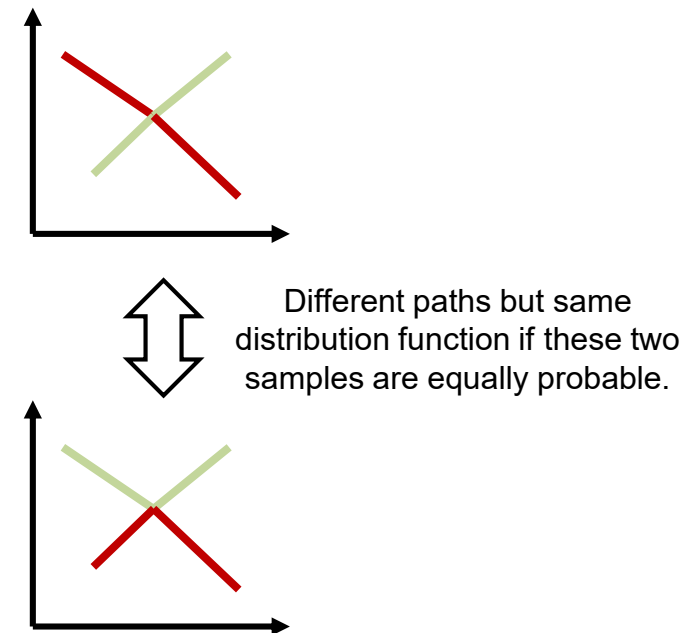
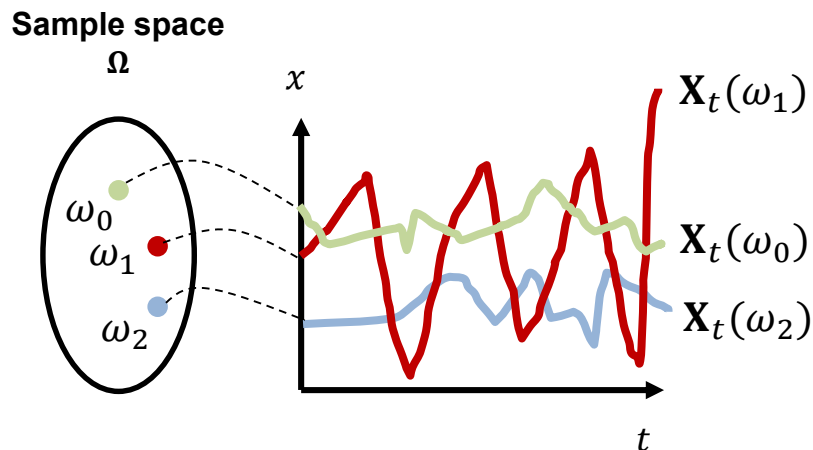


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Preliminaries on SDE (1/5)

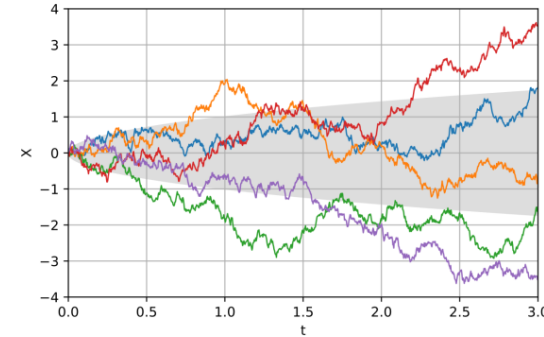
- Stochastic process
 - A parameterized collection of random variables $\{\mathbf{X}_t\}$
 - A path of $\mathbf{X}_t$ with a fixed sample ω_0 is a sample function: $t \rightarrow \mathbf{X}_t(\omega_0)$
 - Several versions of paths can form the same probability distribution, and we prefer continuous ones





Preliminaries on SDE (2/5)

- Markov process
 - Conditioned on the present state, its past and future are statistically independent.
- Diffusion process
 - A Markov process that has continuous paths.
- Brownian motion
 - A Markov process that has the following properties:
 - $W_0 = 0$ (for simplicity)
 - W_t is almost surely continuous (Wiener process is the continuous version)
 - W_t has independent and stationary increments
 - $W_t - W_s \sim \mathcal{N}(0, t - s)$, $0 \leq s \leq t$
 - Expectations
 - $\mathbf{E}[(W_t - W_s)^2] = t - s$, $0 \leq s \leq t$; $\mathbf{E}[W_s \cdot W_t] = \min(s, t)$
 - Quadratic variation $[W_t, W_t]$ is a random process but behaves as almost surely deterministic as t



$$[W_t, W_t] \triangleq \lim_{\Delta t_k \rightarrow 0} \sum_{t_k=0}^t (W_{t_{k+1}} - W_{t_k})^2 \text{ where } \Delta t_k = t_{k+1} - t_k$$

$$\text{Var}[[W_t, W_t] - t] \xrightarrow{\Delta t_k \rightarrow 0} 0 \Rightarrow [W_t, W_t] = t \text{ (almost surely)} \Rightarrow \underline{(dW_t)^2 = dt \text{ (almost surely)}}$$

where SDE differs from ODE

- Reference textbooks
 - G. A. Pavliotis, Stochastic Processes and Applications, Springer, 2014. (**Chap1-4 recommended**)
 - B. Øksendal, Stochastic Differential Equations, 6th ed., Springer, 2003. (mathematical rigor)



Preliminaries on SDE (3/5)

- Forward SDE:

$$d\mathbf{x}_t = \overset{\text{drift}}{\mathbf{f}(\mathbf{x}_t, t)}dt + \overset{\text{diffusion}}{g_t}d\mathbf{w}_t$$

- Drift coefficient $\mathbf{f}(\mathbf{x}, t)$ controls the behavior of mean $\mathbf{E}[\mathbf{x}_t - \mathbf{x}_s]$
- Diffusion coefficient $g(t)$ controls the behavior of variance $\mathbf{E}[\|\mathbf{x}_t - \mathbf{x}_s\|^2]$ with Wiener process $\mathbf{w}_t$

- Solver example: Euler-Maruyama method (1st-order)

$$\Delta\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)\Delta t + g_t\Delta\mathbf{w}_t, \Delta\mathbf{w}_t \sim \mathcal{N}(0, \Delta t)$$

- Connection to forward DPM $q(\mathbf{x}_t|\mathbf{x}_0) = N(\alpha_t\mathbf{x}_0, \sigma_t^2 I)$
 - α_t defined by drift coefficient $\mathbf{f}(\mathbf{x}, t)$
 - σ_t^2 defined by α_t and diffusion coefficient g_t



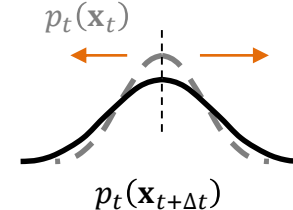
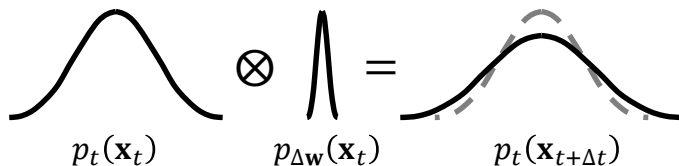
Preliminaries on SDE (4/5)

- Equivalent forward ODE:

- Drift with $-\frac{1}{2} g_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ has the same effect as diffusion with $g_t d\mathbf{w}_t$

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt + g_t d\mathbf{w}_t \quad \longleftrightarrow \quad d\mathbf{x}_t = \left(\mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2} g_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \right) dt$$

same $p_t(\mathbf{x}_t)$



- Reverse ODE:

- Let reverse time $\tau = T - t \therefore d\tau = -dt$

$$d\mathbf{x}_\tau = \left(-\mathbf{f}(\mathbf{x}_\tau, \tau) + \frac{1}{2} g_\tau^2 \nabla_{\mathbf{x}_\tau} \log p_\tau(\mathbf{x}_\tau) \right) d\tau$$



Preliminaries on SDE (5/5)

- Equivalent reverse SDE:
 - $\tilde{\mathbf{w}}_\tau$ is a reverse-time Wiener process

$$d\mathbf{x}_\tau = \left(-\mathbf{f}(\mathbf{x}_\tau, \tau) + \frac{1}{2} g_\tau^2 \nabla_{\mathbf{x}_\tau} \log p_\tau(\mathbf{x}_\tau) \right) d\tau$$

$$\Updownarrow \text{ same } p_\tau(\mathbf{x}_\tau)$$

$$d\mathbf{x}_\tau = \left(-\mathbf{f}(\mathbf{x}_\tau, \tau) + g_\tau^2 \nabla_{\mathbf{x}_\tau} \log p_\tau(\mathbf{x}_\tau) \right) d\tau + g_\tau d\tilde{\mathbf{w}}_\tau$$

$$\Updownarrow t = T - \tau$$

$$\boxed{d\mathbf{x}_t = \left(\mathbf{f}(\mathbf{x}_t, t) - g_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \right) dt + g_t d\tilde{\mathbf{w}}_t}$$

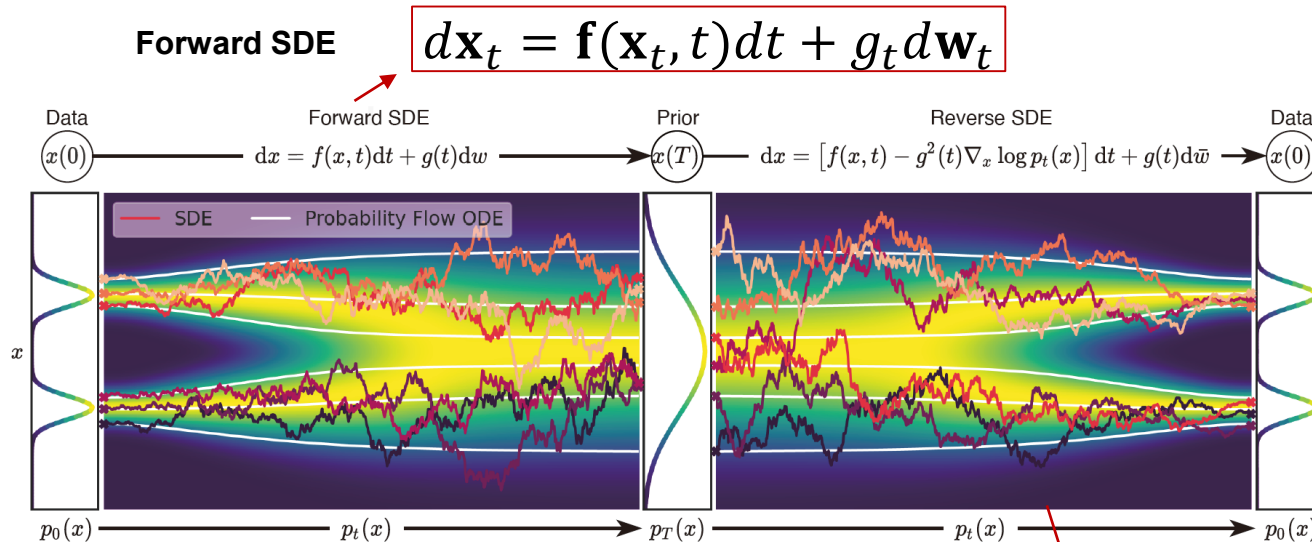


Score matching is required for solving reverse SDE



Score-based generative modeling through SDE (ICLR'21) (1/3)

- SDE formulation for diffusion model
 - Reverse SDE needs the score function on $\mathbf{x}_t$
 - Applies the same score matching in SMLD: use $\mathbf{s}_\theta(\mathbf{x}, t)$ to learn $\nabla_{\mathbf{x}_i} \log q(\mathbf{x}_i)$ or $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ (Note that $p_t(x) = q_t(x)$ for SDE)
 - $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \approx \mathbf{s}_\theta(\mathbf{x}, t) = -\epsilon_\theta(\mathbf{x}, t)/\sigma_t$



Reverse SDE

$$d\mathbf{x}_t = (\mathbf{f}(\mathbf{x}_t, t) - g_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t))dt + g_t d\tilde{\mathbf{w}}_t$$



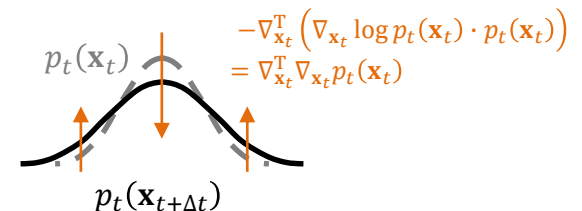
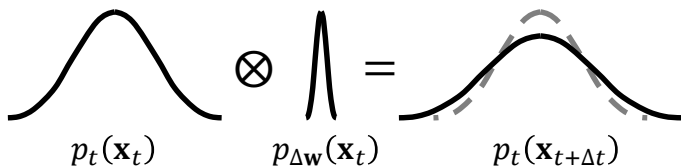
Score-based generative modeling through SDE (ICLR'21) (2/3)

- Contributions of this paper
 - Unify SMLD and DDPM in the same framework and enable infinite alternatives (with different drift and diffusion coefficients), including
 - Variance-preservative (VP) SDE (similar to DDPM)
 - Variance-explosive SDE (similar to SMLD)
 - Sub-VP SDE: new
 - Bring out the ODE version of the process for deterministic process
 - Stochastic Gaussian diffusion $g_t d\mathbf{w}_t$ equivalent to deterministic drift $-\frac{1}{2} g_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$
 - Reverse ODE for acceleration
 - Forward ODE for unique encoding
 - Exact likelihood evaluation

Forward/reverse ODE

$$d\mathbf{x}_t = \mathbf{f}(\mathbf{x}_t, t)dt + g_t d\mathbf{w}_t \quad \longleftrightarrow \quad d\mathbf{x}_t = \left(\mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2} g_t^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) \right) dt$$

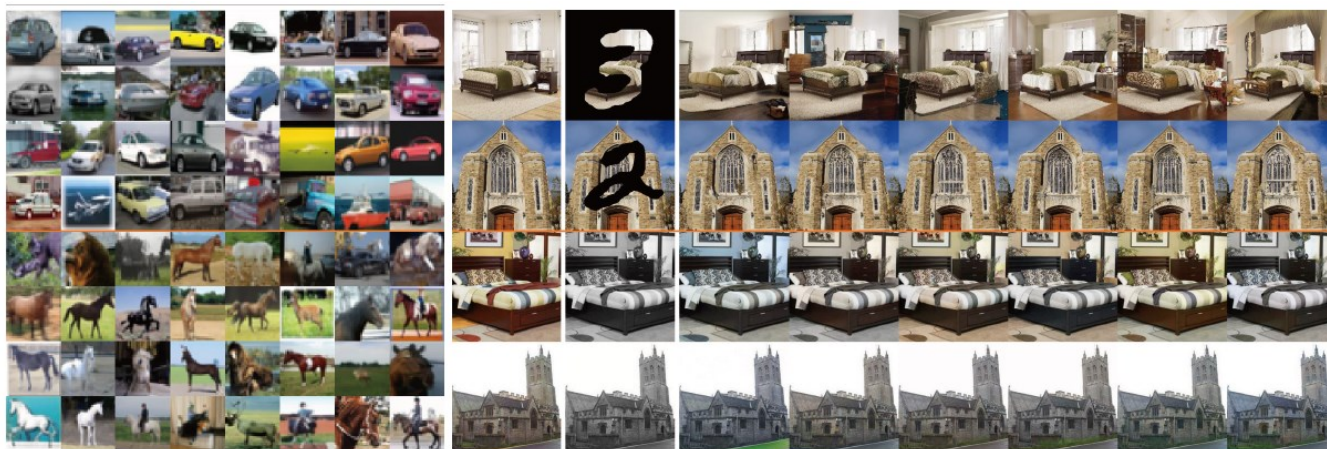
same $p_t(\mathbf{x}_t)$



Score-based generative modeling through SDE (ICLR'21) (3/3)

- Contributions of this paper (cont'd)
 - Solve SDE with numerical predictor-corrector samplers
 - Predictor: generate x_i^p using the previous sample x_{i-1} ; different methods include:
 - Ancestral sampling: Markov process for ELBO minimization; $p_\theta(x_{t-1}|x_t)$ in form of $q(x_{t-1}|x_t, x_0)$
 - Reverse diffusion: direct discretization on time (Euler-Maruyama 1st order approximation of SDE)
 - Probability flow ODE: step-adjustable from black-box ODE solver (VP-ODE links to DDIM)
 - Corrector: correct the marginal distribution of x_i from x_i^p ; apply:
 - Langevin dynamics
 - Ease conditional generation with Bayes' rule

$$dx = \{f(x, t) - g(t)^2[\nabla_x \log p_t(x) + \nabla_x \log p_t(y | x)]\}dt + g(t)d\bar{w}$$



Car-conditioned

Inpainting

Horse-conditioned

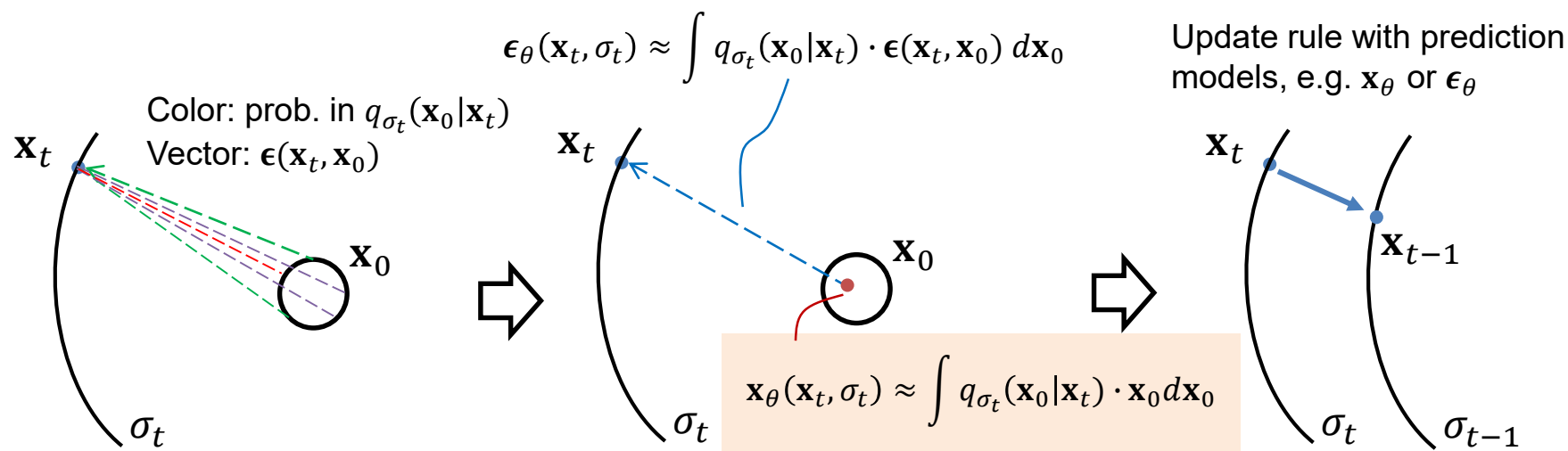
Colorization

Implication from SDE

- Reverse process equals to solve SDE/ODE with prediction models, e.g. ODE solvers for

$$d\mathbf{x}_t = \left(\mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2} g_t^2 \mathbf{s}_\theta \right) dt = \left(\mathbf{f}(\mathbf{x}_t, t) + \frac{1}{2} \frac{g_t^2}{\sigma_t} \boldsymbol{\epsilon}_\theta \right) dt$$

- Updates with conditional predictions





Outline

- **Method**
 - DDPM/DDIM
 - SMLD
 - SDE/ODE
 - **VDM**
 - **EDM**
 - FM
 - Appendices for equations



Variational diffusion model (VDM) (1/2)

- Extend likelihood-based optimization with variational lower bound to continuous-time model
 - Reverse process as DDPM with $p_\theta(x_{t-1}|x_t)$ in form of $q(x_{t-1}|x_t, x_0)$
- SNR-based discrete-time model (T-step; DDPM likelihood) $q(\mathbf{z}_t|\mathbf{x}) = \mathcal{N}(\alpha_t \mathbf{x}, \sigma_t^2 \mathbf{I})$

Data prediction model

$$\mathcal{L}_T(\mathbf{x}) = \frac{T}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), i \sim U\{1, T\}} [(\text{SNR}(s) - \text{SNR}(t)) \|\mathbf{x} - \hat{\mathbf{x}}_\theta(\mathbf{z}_t; t)\|_2^2] \quad \text{SNR}(t) = \alpha_t^2 / \sigma_t^2$$

Noise prediction model

$$\mathcal{L}_T(\mathbf{x}) = \frac{T}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), i \sim U\{1, T\}} [(\exp(\gamma_\eta(t) - \gamma_\eta(s)) - 1) \|\epsilon - \hat{\epsilon}_\theta(\mathbf{z}_t; t)\|_2^2] \quad \text{SNR}(t) = \exp(-\gamma_\eta(t))$$

- Continuous-time model ($T \rightarrow \infty$)
 - Training with continuous time steps for t

Data prediction model

$$\mathcal{L}_\infty(\mathbf{x}) = -\frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I}), t \sim \mathcal{U}(0, 1)} [\text{SNR}'(t) \|\mathbf{x} - \hat{\mathbf{x}}_\theta(\mathbf{z}_t; t)\|_2^2]$$

$\downarrow \quad v \equiv \text{SNR}(t)$

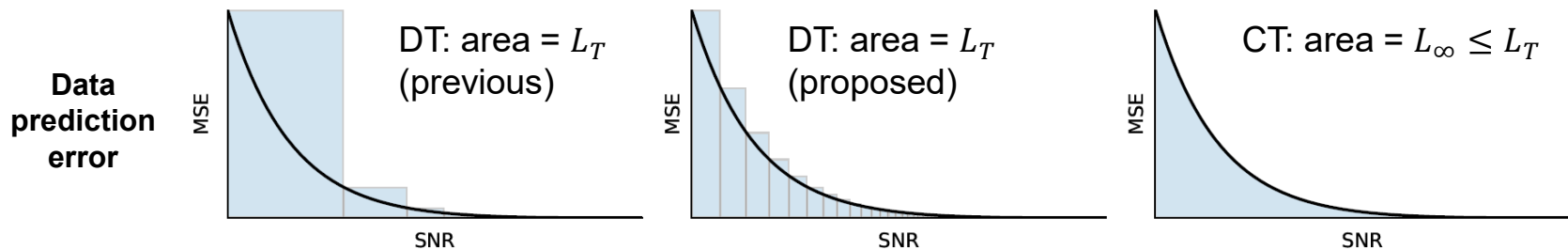
$$\mathcal{L}_\infty(\mathbf{x}) = \frac{1}{2} \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbf{I})} \int_{\text{SNR}_{\min}}^{\text{SNR}_{\max}} \|\mathbf{x} - \tilde{\mathbf{x}}_\theta(\mathbf{z}_v, v)\|_2^2 dv$$

Continuous DDPM likelihood only relates to $\text{SNR}_{\min}$ and $\text{SNR}_{\max}$, regardless of $\text{SNR}(t)$ shape (scheduling), for data prediction



Variational diffusion model (VDM) (2/2)

- Contributions of this paper
 - Generalize likelihood optimization to continuous-time model
 - Reason discrete-time (DT) DDPM model as **numeric-integration upper bound** of continuous-time (CT) one for likelihood evaluation



- Suggest SNR-based scheduling, instead of time-based
 - Assume SNR is monotonically decreasing
- Show CT DDPM generation converges to reverse SDE

$$d\mathbf{z}_t = [\underline{f(t)\mathbf{z}_t} - g^2(t)s_\theta(\mathbf{z}_t; t)]dt + g(t)d\mathbf{W}_t$$

Specific drift formulation for diffusion model: linear on $\mathbf{z}_t \Rightarrow \mathbf{f}(\mathbf{x}_t, t) = f_t \mathbf{x}_t$

- Optimize true likelihood for DM (not L_{simple}) and outperform AR models
 - Propose learnable scheduling $\gamma_\eta(t)$, including $\text{SNR}_{\min}$ and $\text{SNR}_{\max}$
 - Observe that fidelity weights more on coarse/low-SNR features but likelihood on detailed/high-SNR ones $\Rightarrow$ likelihood-optimized scheduling shifts SNR to larger values



Elucidating design space of DMs (EDM) (1/2)

- Convert reverse SDE into a simple form of ODE and then solve with handy ODE solvers

$$dx_t = (f_t x_t - g_t^2 s_\theta) dt + g_t dw_t \Rightarrow dx_t = \left(f_t x_t - \frac{g_t^2 s_\theta}{2} \right) dt$$

Can be solved exactly
without discretization errors

$$\underline{(\tilde{x}_t = x_t / \alpha_t, \alpha_t = e^{\int_0^t f_s ds})} \Rightarrow d\tilde{x}_t = - \frac{g_t^2 s_\theta}{2\alpha_t} dt$$

velocity

VE form: $\tilde{x}_t = x_0 + \frac{\sigma_t}{\alpha_t} \epsilon_t$

Solved by 2nd-order
Heun's method;
two neural function
evaluations (**NFEs**)
per update.

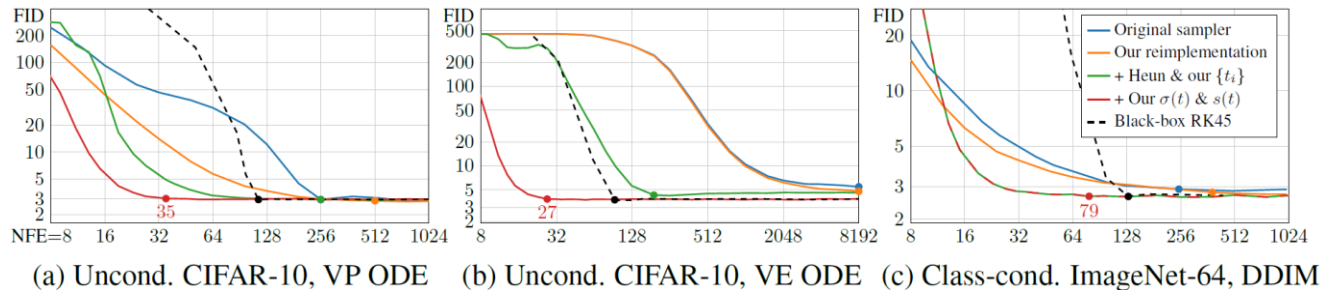
$y'(t) = f(t, y(t)),$	$y(t_0) = y_0,$
$\tilde{y}_{i+1} = y_i + hf(t_i, y_i)$	
predictor	
$y_{i+1} = y_i + \frac{h}{2}[f(t_i, y_i) + f(t_{i+1}, \tilde{y}_{i+1})],$	
corrector	

Credit: Wikipedia.



Elucidating design space of DMs (EDM) (2/2)

- Contributions of this paper
 - Propose to use high-order ODE solvers for DM generation
 - Reparameterization for eliminating the linear drift term for ODE solving
 - Better FID for medium-size NFE
 - Karras scheduler** to allocate more steps near large SNR



- Identify SDE family and improve diversity by replacing some truncation errors with forward-process stochasticity
 - The actual proposed stochastic sampler uses a travel-back technique, which is not an SDE solver but more accurate than Euler-Maruyama method

$$dx_t = \left(f_t x_t - \frac{g_t^2 s_\theta}{2} - r_t \frac{g_t^2 s_\theta}{2} \right) dt + r_t g_t dw_t \Rightarrow d\tilde{x}_t = \underbrace{-(1 + r_t) \frac{g_t^2 s_\theta}{2\alpha_t}}_{\text{Substrate more estimated noise}} dt + \underbrace{r_t \frac{g_t}{\alpha_t} dw_t}_{\text{Add more stochastic noise}}$$

Family of reverse SDE

Substrate more estimated noise Add more stochastic noise



Implication from SDE/ODE solver

- Only prediction term needs approximation

$$d\tilde{x}_t = -(1+r)\frac{g_t^2 s_\theta}{2\alpha_t} dt + r\frac{g_t}{\alpha_t} dw_t \longrightarrow \underbrace{\tilde{x}_s = \tilde{x}_t}_{\text{Exact term } \mathbf{E}} - (1+r)\underbrace{\int_t^s \frac{g_\tau^2 s_\theta}{2\alpha_\tau} d\tau}_{\text{Prediction term } \mathbf{N} \text{ (approximated)}} + r\underbrace{\int_t^s \frac{g_\tau}{\alpha_\tau} dw_\tau}_{\text{Wiener term } \mathbf{W} \text{ (exact)}}$$

- Evaluation of prediction term depends on the choice of
 - Prediction type, e.g. data or noise prediction (source of non-linearity)
 - Discretization variable, e.g. time, log-SNR $\lambda = \log \frac{\alpha_t}{\sigma_t}$, etc.
 - Solver, e.g. 1st/2nd/3rd-order, single-/multi-step, etc.
 - Predictor-corrector scheme, e.g. P only or PC

$$\begin{aligned} dt &= \frac{dt}{d\lambda} d\lambda \\ \downarrow \\ \Delta t &\neq \frac{dt}{d\lambda} \Delta \lambda \end{aligned}$$

- Implication
 - DDIM equals to 1st-order predictor on log-SNR for data/noise prediction
 - DDPM equals to 1st-order predictor on log-SNR for data prediction
 - Large-step DDPM does not handle the correlation between N and W and thus degrades sampling quality



Summary for Continuous-time SDE/ODE

- One forward SDE with specific formulation of $d\mathbf{x}_t = f_t\mathbf{x}_t dt + g_t d\mathbf{w}_t$ corresponds to a family of forward and reverse SDEs
 - Many different solvers can be used
 - (can be easily integrated with classifier-free guidance)
- Different prediction networks are interchangeable
 - Only if convertible to score function
 - SDE is not likelihood-driven but score-driven
- Many training hyperparameters need to be set up (same as DPM)
 - Scheduling of α_t, σ_t ; reweighting w_t ; prediction type
- ODE is an important variant
 - Forward ODE for unique encoding (**invertible**)
 - Reverse ODE for fast inference (reasonable)



Outline

- **Method**
 - DDPM/DDIM
 - SMLD
 - SDE/ODE
 - **FM**
 - Appendices for equations



Preliminaries on FM (1/2)

- Deterministic coupling
 - Given r.v. $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ with $X \sim \mu$ and $Y \sim \nu$
 - A coupling is $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ with a joint probability $\pi(X, Y)$
 - A coupling is deterministic if there exists a measurable function $T: \mathcal{X} \rightarrow \mathcal{Y}$ such that $Y = T(X)$
 - T is also called the transport map
- Optimal transport/coupling
 - For a cost function $c(x, y)$ on $\mathcal{X} \times \mathcal{Y}$, the optimal coupling minimizes $\mathbb{E}_{\pi}[c(X, Y)]$ and such $\pi(X, Y)$ is called optimal transference plan
 - Note
 - WGAN finds optimal coupling for L1 cost
 - FID considers optimal coupling for L2 cost
 - For μ and ν both Gaussians, the optimal coupling for L2 cost can be achieved by a linear transformation T



Preliminaries on FM (2/2)

- Continuous normalizing flow (CNF)
 - Probability density path $p: [0,1] \times \mathbb{R}^d \rightarrow \mathbb{R}$; i.e. $p_t(\mathbf{z}_t)$
 - A flow $\phi: [0,1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$, i.e. $\mathbf{z}_t = \phi_t(\mathbf{x}_0)$ as a time-dependent transport map; note that $\mathbf{z}_0 = \mathbf{x}_0$, but $\mathbf{z}_1$ may be only close to $\mathbf{x}_1$
 - CNF models a velocity field $\mathbf{v}: [0,1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ via the ODE:

$$\frac{d}{dt}\phi_t(x) = v_t(\phi_t(x))$$

Here $x = \mathbf{x}_0$

$$\phi_0(x) = x$$

- Mass conservation formula $\frac{\partial}{\partial t} p_t(\mathbf{z}_t) = \nabla_{\mathbf{z}_t} \cdot (p_t(\mathbf{z}_t) \cdot \mathbf{v}_t(\mathbf{z}_t))$
- Connection to SDE for DM
 - Velocity field v equals to the drift coefficient of the equivalent ODE for DM
 - CNF directly models velocity $v_\theta = f_t x_t - \frac{g_t^2 s_\theta}{2}$, not score s_θ
 - Mass conservation formula exactly serves the formula of the SDE drift coefficient



Flow matching (FM, ICLR'23) (1/4)

- Propose to match the flow directly via the L2 FM objective

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, p_t(x)} \left\| \underbrace{v_t(x)}_{\text{Learnable velocity model}} - \underbrace{u_t(x)}_{\text{Target vector field (defined by scheduling)}} \right\|^2$$

In this slide, $x = \mathbf{z}_t$

- Implement via conditional FM matching since u_t is not tractable
 - Similar to conditional noise/data/score matching
 - Note: in this paper, time $t=0$ for pure noise dist. and $t=1$ for clean data dist.

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, q(x_1), p_t(x|x_1)} \left\| v_t(x) - \underbrace{u_t(x|x_1)}_{\text{Conditional target vector field}} \right\|^2$$

Conditional target vector field

- Show that marginal velocity field equals to the expected value of the conditional velocity field

$$u_t(x) = \int u_t(x|x_1) \frac{p_t(x|x_1)q(x_1)}{p_t(x)} dx_1 = \frac{p(\mathbf{z}_t, \mathbf{x}_1)}{p(\mathbf{z}_t)} = p(\mathbf{x}_1|\mathbf{z}_t)$$



Flow matching (FM, ICLR'23) (2/4)

- Consider data-dependent Gaussians for conditional probability path
 - μ_t and σ_t depend on $\mathbf{x}_1$

$$p_t(x|x_1) = \mathcal{N}(x \mid \mu_t(x_1), \sigma_t(x_1)^2 I)$$

- Suggest the simplest conditional flow (from infinite alternatives)
 - Conditional flow ψ_t decided by μ_t and σ_t
 - $\mathbf{z}_t$ determined by $\mathbf{x}_0$ and $\mathbf{x}_1$ (together define the whole forward/reverse process)

$$\psi_t(x) = \sigma_t(x_1)x + \mu_t(x_1) \Rightarrow \mathbf{z}_t = \sigma_t(\mathbf{x}_1)\mathbf{x}_0 + \boldsymbol{\mu}_t(\mathbf{x}_1), \mathbf{x}_0 \sim \mathcal{N}(0,1)$$

- Solve the conditional target vector field in a closed form
 - Conditional vector field $u_t(\mathbf{z}_t|\mathbf{x}_1)$ determined by μ_t , σ_t , $\mathbf{x}_0$ and $\mathbf{x}_1$

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)|x_1) \Rightarrow u_t(\mathbf{z}_t|\mathbf{x}_1) = \sigma'_t(\mathbf{x}_1)\mathbf{x}_0 + \boldsymbol{\mu}'_t(\mathbf{x}_1)$$



Flow matching (FM, ICLR'23) (3/4)

- Explore examples of vector fields, include
 - Conventional VP and VE diffusion schedules, leading to their ODE forms
 - Natural choice of linear μ_t and σ_t , leading to conditional optimal transport (for Gaussians), fewer reverse steps, and faster reverse sampling

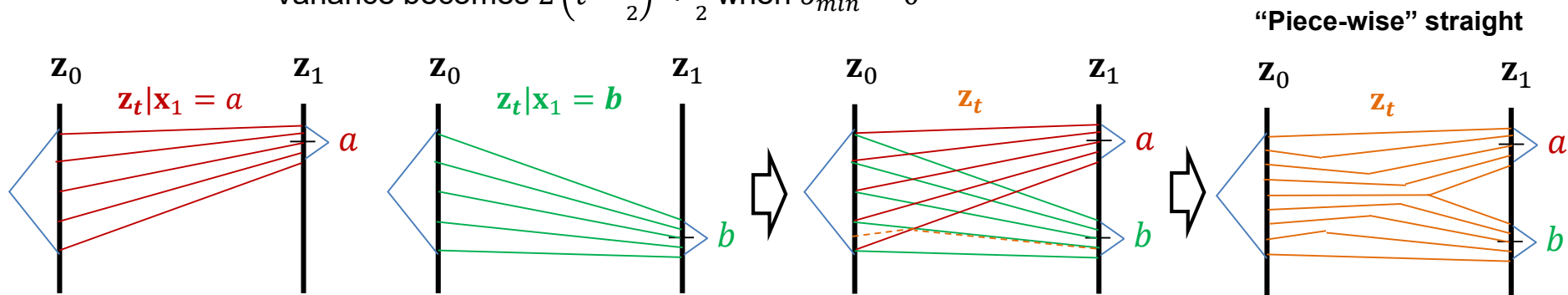
$$\mu_t(x) = tx_1, \text{ and } \sigma_t(x) = 1 - (1 - \sigma_{\min})t$$

$$\psi_t(x) = (1 - (1 - \sigma_{\min})t)x + tx_1$$

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,q(x_1),p(x_0)} \left\| v_t(\psi_t(x_0)) - \left(x_1 - (1 - \sigma_{\min})x_0 \right) \right\|^2$$

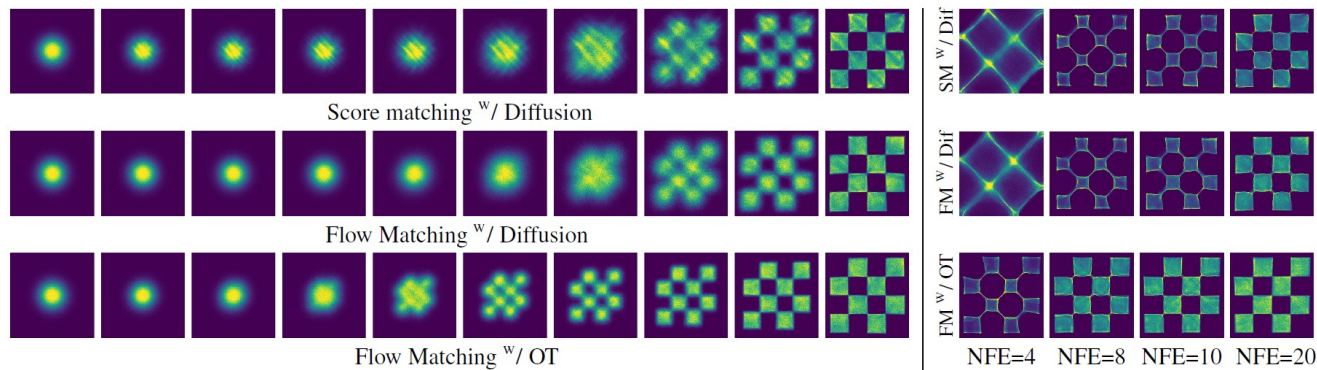
$u_t(\mathbf{z}_t|\mathbf{x}_1)$: data – noise; constant/straight

- Same as straight rectified flow [ICLR'23] when $\sigma_{\min} = 0$
- Variance becomes $2\left(t - \frac{1}{2}\right)^2 + \frac{1}{2}$ when $\sigma_{\min} = 0$

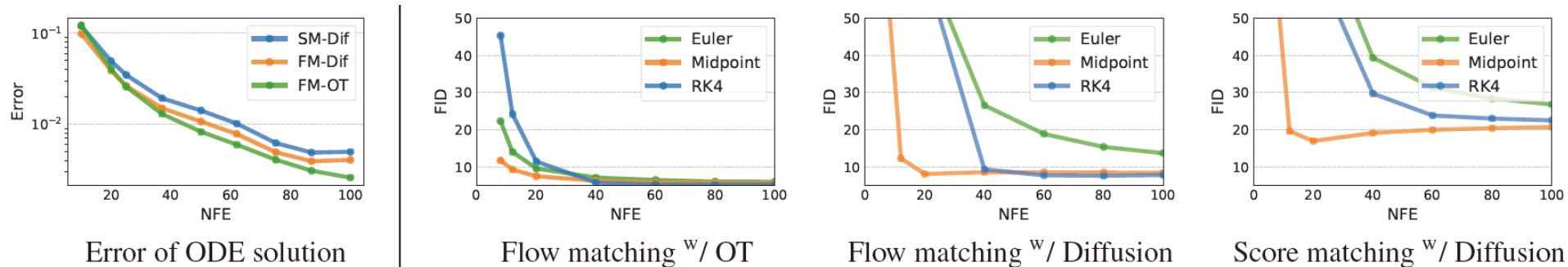


Flow matching (FM, ICLR'23) (4/4)

- Trajectories over different training objectives
 - FM w/ OT introduces checkerboard patterns much earlier



- Fewer evaluations for similar numerical error and sample quality (ImageNet 32x32)
 - FM w/ OT also has much smaller performance gap between Euler and Heun's





Summary for FM

- Invertible forward/reverse process defined by specific conditional formulation of $\mathbf{z}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$
 - Simply match $\mathbf{v}_t(\mathbf{z}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1; \theta)$ to $\mathbf{x}_1 - \mathbf{x}_0$ for each noise/clean pair $(\mathbf{x}_0, \mathbf{x}_1)$
 - Simple solvers are sufficient
 - General case: $\mathbf{z}_t = \sigma_t(\mathbf{x}_1)\mathbf{x}_0 + \boldsymbol{\mu}_t(\mathbf{x}_1)$
- Velocity prediction network is all you need
 - Convertible to other predictions via ODE formulation
 - FM is not likelihood-driven but flow-driven
- Much fewer hyperparameters need to be set up
 - Reweighting w_t (fix α_t , σ_t , and prediction type)
- FM is becoming the mainstream
 - Keeps all benefits of ODE
 - Provides faster inference



Outline

- **Method**
 - DDPM/DDIM
 - SMLD
 - SDE/ODE
 - FM
 - **Appendices for equations**
 - Noise schedule
 - Diffusion process
 - Wiener process
 - Ito process
 - SDE for DM



$$\alpha_t = s(t)$$

$$\sigma_t = s(t)\sigma(t)$$

Examples of noise schedule

	VP [49]	VE [49]	iDDPM [37] + DDIM [47]	Ours (“EDM”)
Sampling (Section 3)				
ODE solver	Euler	Euler	Euler	2 nd order Heun
Time steps	$t_{i < N} \quad 1 + \frac{i}{N-1}(\epsilon_s - 1)$	$\sigma_{\max}^2 (\sigma_{\min}^2 / \sigma_{\max}^2)^{\frac{i}{N-1}}$	$u_{\lfloor j_0 + \frac{M-1-j_0}{N-1}i + \frac{1}{2} \rfloor}$, where $u_M = 0$ $u_{j-1} = \sqrt{\frac{u_j^2 + 1}{\max(\bar{\alpha}_{j-1}/\bar{\alpha}_j, C_1)}} - 1$	$(\sigma_{\max}^{\frac{1}{\rho}} + \frac{i}{N-1}(\sigma_{\min}^{\frac{1}{\rho}} - \sigma_{\max}^{\frac{1}{\rho}}))^\rho$
β-linear, e.g. DDPM, LDM				
Schedule	$\sigma(t)$	$\sqrt{t}$	t	t
Scaling	$s(t)$	1	1	1
Network and preconditioning (Section 5)				
Architecture of F_θ	DDPM++	NCSN++	DDPM	(any)
Skip scaling $c_{\text{skip}}(\sigma)$	1	1	1	$\sigma_{\text{data}}^2 / (\sigma^2 + \sigma_{\text{data}}^2)$
Output scaling $c_{\text{out}}(\sigma)$	$-\sigma$	σ	$-\sigma$	$\sigma \cdot \sigma_{\text{data}} / \sqrt{\sigma_{\text{data}}^2 + \sigma^2}$
Input scaling $c_{\text{in}}(\sigma)$	$1/\sqrt{\sigma^2 + 1}$	1	$1/\sqrt{\sigma^2 + 1}$	$1/\sqrt{\sigma^2 + \sigma_{\text{data}}^2}$
Noise cond. $c_{\text{noise}}(\sigma)$	$(M-1)\sigma^{-1}(\sigma)$	$\ln(\frac{1}{2}\sigma)$	$M-1 - \arg \min_j u_j - \sigma $	$\frac{1}{4} \ln(\sigma)$
Training (Section 5)				
Noise distribution	$\sigma^{-1}(\sigma) \sim \mathcal{U}(\epsilon_t, 1)$	$\ln(\sigma) \sim \mathcal{U}(\ln(\sigma_{\min}), \ln(\sigma_{\max}))$	$\sigma = u_j, j \sim \mathcal{U}\{0, M-1\}$	$\ln(\sigma) \sim \mathcal{N}(P_{\text{mean}}, P_{\text{std}}^2)$
Loss weighting $\lambda(\sigma)$	$1/\sigma^2$	$1/\sigma^2$	$1/\sigma^2$ (note: *)	$(\sigma^2 + \sigma_{\text{data}}^2) / (\sigma \cdot \sigma_{\text{data}})^2$
Parameters				
	$\beta_d = 19.9, \beta_{\min} = 0.1$	$\sigma_{\min} = 0.02$	$\bar{\alpha}_j = \sin^2(\frac{\pi}{2} \frac{j}{M(C_2+1)})$	$\sigma_{\min} = 0.002, \sigma_{\max} = 80$
	$\epsilon_s = 10^{-3}, \epsilon_t = 10^{-5}$	$\sigma_{\max} = 100$	$C_1 = 0.001, C_2 = 0.008$	$\sigma_{\text{data}} = 0.5, \rho = 7$
	$M = 1000$		$M = 1000, j_0 = 8^\dagger$	$P_{\text{mean}} = -1.2, P_{\text{std}} = 1.2$

* iDDPM also employs a second loss term L_{vib}

[†] In our tests, $j_0 = 8$ yielded better FID than $j_0 = 0$ used by iDDPM

SNR α_t/σ_t range [0.0066:95.37] [0.01:50] [0.0125:500]

Credit: Karras et al., NeurIPS'22; EDM.

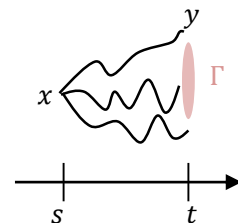


Diffusion process (1/3)

- Formal definition of diffusion process with transition function:

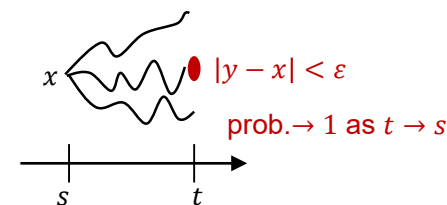
- Transition function $p(\Gamma, t|x, s)$

$$p(\Gamma, t|x, s) := P[X_t \in \Gamma | X_s = x] = \int_{\Gamma} p(dy, t|x, s) \stackrel{\text{if } x, y \text{ continuous}}{\downarrow} \int_{\Gamma} p(y, t|x, s) dy$$



- Continuity (almost surely): continuous sample paths

$$\forall \varepsilon > 0, \int_{|y-x|>\varepsilon} p(dy, t|x, s) = o(t-s)$$



- Drift coefficient $b(x, s)$

$$\exists b(x, s) \text{ s.t. } \int_{|y-x|<\varepsilon} (y-x) p(dy, t|x, s) = b(x, s)(t-s) + o(t-s)$$

$$\Rightarrow b(x, s) = \lim_{t \rightarrow s} \mathbf{E} \left[\frac{X_t - X_s}{t-s} | X_s = x \right] \quad (\text{Changing rate of the mean})$$

- Diffusion coefficient $\Sigma(x, s)$

$$\exists \Sigma(x, s) \text{ s.t. } \int_{|y-x|<\varepsilon} (y-x)^2 p(dy, t|x, s) = \Sigma(x, s)(t-s) + o(t-s)$$

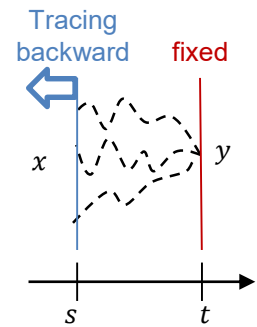
$$\Rightarrow \Sigma(x, s) = \lim_{t \rightarrow s} \mathbf{E} \left[\frac{|X_t - X_s|^2}{t-s} | X_s = x \right] \quad (\text{Changing rate of the variance})$$



Diffusion process (2/3)

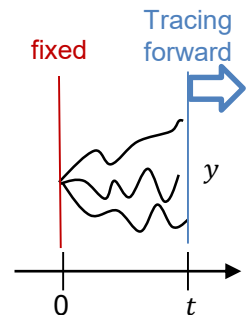
- Kolmogorov backward equation (KBE)
 - On transition probability density $p(y, t|x, s)$
 - Final value problem with $p(y, t|x = y, s = t) = \delta(x - y)$

$$-\frac{\partial p(y, t|x, s)}{\partial s} = \underset{\substack{\uparrow \\ \text{change for } s \text{ tracing backward}}}{b(x, s)} \frac{\partial p(y, t|x, s)}{\partial x} + \frac{1}{2} \underset{\substack{\uparrow \\ \text{Drift and diffusion over } x \text{ for the current } s}}{\Sigma(x, s)} \frac{\partial^2 p(y, t|x, s)}{\partial x^2}$$



- Kolmogorov forward equation (KFE)/Fokker-Planck equation
 - On probability density $p(y, t)$
 - Initial value problem with $p(y, t = 0) = \rho(y)$

$$\frac{\partial p(y, t)}{\partial t} = -\frac{\partial}{\partial y} (b(y, t)p(y, t)) + \frac{1}{2} \frac{\partial^2}{\partial y^2} (\Sigma(y, t)p(y, t))$$



- Multi-dimensional extension

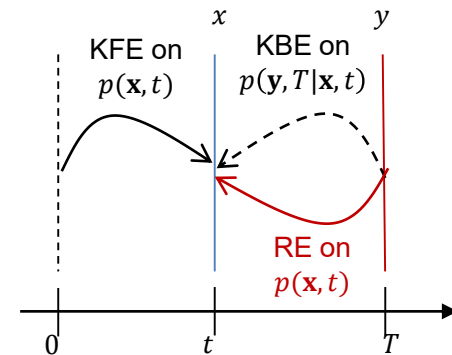
$$-\frac{\partial p(\mathbf{y}, t|\mathbf{x}, s)}{\partial s} = \underset{\substack{\uparrow \\ \mathbb{R}^n}}{\mathbf{b}(\mathbf{x}, s)} \cdot \underset{\substack{\uparrow \\ \text{Vector inner product}}}{\nabla_{\mathbf{x}}} p(\mathbf{y}, t|\mathbf{x}, s) + \frac{1}{2} \underset{\substack{\uparrow \\ \mathbb{R}^n \times \mathbb{R}^n}}{\Sigma(\mathbf{x}, s)} : \underset{\substack{\uparrow \\ \text{Frobenius inner product}}}{(\nabla_{\mathbf{x}} \nabla_{\mathbf{x}}^T p(\mathbf{y}, t|\mathbf{x}, s))}$$

$$\begin{aligned} -\frac{\partial p(\mathbf{y}, t)}{\partial t} &= -\nabla_{\mathbf{y}} \cdot (\mathbf{b}(\mathbf{y}, t)p(\mathbf{y}, t)) + \frac{1}{2} (\nabla_{\mathbf{y}} \nabla_{\mathbf{y}}^T) : (\Sigma(\mathbf{y}, t)p(\mathbf{y}, t)) \\ &= -\nabla_{\mathbf{y}}^T (\mathbf{b}(\mathbf{y}, t)p(\mathbf{y}, t)) + \frac{1}{2} \nabla_{\mathbf{y}}^T (\Sigma(\mathbf{y}, t)p(\mathbf{y}, t)) \nabla_{\mathbf{y}} \quad (\text{matrix-vector form}) \end{aligned}$$



Diffusion process (3/3)

- Reverse-time equation (RE) [Anderson, 1982]
 - Start from the joint distribution $p(\mathbf{x}, t, \mathbf{y}, T) = p(\mathbf{y}, T | \mathbf{x}, t) p(\mathbf{x}, t)$
 - Then calculate $\frac{\partial p(\mathbf{x}, t, \mathbf{y}, T)}{\partial t}$ by
 - Applying KBE on $p(\mathbf{y}, T | \mathbf{x}, t)$ for having $\frac{\partial p(\mathbf{y}, T | \mathbf{x}, t)}{\partial t}$
 - Applying KFE on $p(\mathbf{x}, t)$ for having $\frac{\partial p(\mathbf{x}, t)}{\partial t}$
 - Replacing every $p(\mathbf{y}, T | \mathbf{x}, t)$ by $p(\mathbf{x}, t, \mathbf{y}, T) / p(\mathbf{x}, t)$
 - Finally marginalize out $\mathbf{y}$ to have the RE



$$-\frac{\partial p(\mathbf{x}, t)}{\partial t} = -\nabla_{\mathbf{x}}^T \left(-\bar{\mathbf{b}}(\mathbf{x}, t) p(\mathbf{x}, t) \right) + \frac{1}{2} \nabla_{\mathbf{x}}^T \left(\Sigma(\mathbf{x}, t) p(\mathbf{x}, t) \right) \nabla_{\mathbf{x}}$$

$$\bar{\mathbf{b}}(\mathbf{x}, t) = \mathbf{b}(\mathbf{x}, t) - \frac{1}{p(\mathbf{x}, t)} \left((\Sigma(\mathbf{x}, t) p(\mathbf{x}, t)) \nabla_{\mathbf{x}} \right)$$

- Note
 - These PDEs are important for physics and finance, but we will not directly apply them for DM evaluation. Instead, we use them to confirm the equivalence between SDE/ODE and evaluate DM using numerical integration on SDE/ODE.



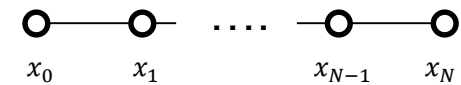
Discrete-time analogy

- KFE and RE are two different viewpoints/evolutions for the same joint distribution over $\{\mathbf{x}_t\}$.

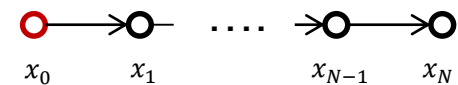
- Consider a discrete-time Markov model

- Joint distribution

$$\begin{aligned} &P(x_0, x_1, \dots, x_{N-1}, x_N) \\ &= P(x_0) \prod_{i=1}^N P(x_i | x_{i-1}, \dots, x_0) \\ &= P(x_0) \prod_{i=1}^N P(x_i | x_{i-1}) \\ &= P(x_N) \prod_{i=N-1}^0 P(x_i | x_{i+1}) \end{aligned}$$



(forward evolution)



(reverse evolution)

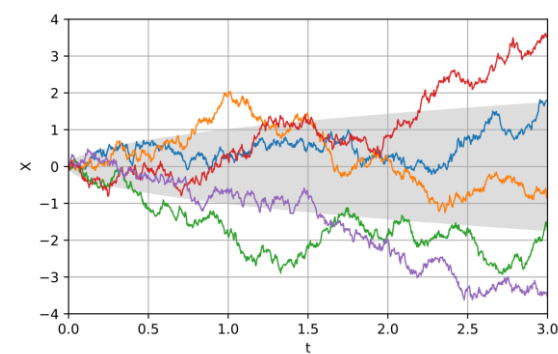
- Note: The variable i is simply an index of r.v. and does not imply causal effect, so does the time index t for DM.



Outline

- **Method**
 - DDPM/DDIM
 - SMLD
 - SDE/ODE
 - FM
 - **Appendices for equations**
 - Noise schedule
 - Diffusion process
 - **Wiener process**
 - **Ito process**
 - **SDE for DM**

Wiener process



Credit: Wikipedia.

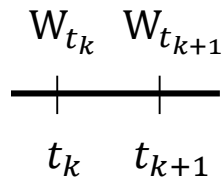
– Characterization

- $W_0 = 0$ (for simplicity)
- W_t is almost surely continuous (Wiener process is the continuous version)
- W_t has independent and stationary increments
- $W_t - W_s \sim \mathcal{N}(0, t - s)$, $0 \leq s \leq t$

– Expectation

- $\mathbf{E}[W_t] = 0$; $\mathbf{E}[W_t^2] = t$
- $\mathbf{E}[(W_t - W_s)^2] = t - s$, $0 \leq s \leq t$
- $\mathbf{E}[W_s \cdot W_t] = \min(s, t)$
- Quadratic variation process: $[W_t, W_t] = t \Rightarrow (dW_t)^2 = dt$ (almost surely, prob. $\rightarrow 1$)

Proof: $[W_t, W_t] \triangleq \lim_{\Delta t_k \rightarrow 0} \sum_{t_k=0}^t (\Delta W_{t_k})^2$ where $\Delta t_k = t_{k+1} - t_k$ and $\Delta W_{t_k} = W_{t_{k+1}} - W_{t_k}$,



$$\begin{aligned}
 \mathbf{E} \left[\left(\sum_{t_k} (\Delta W_{t_k})^2 - t \right)^2 \right] &= \mathbf{E} \left[\sum_{t_i=j} (\Delta W_{t_i})^4 + \sum_{t_i \neq t_j} (\Delta W_{t_i})^2 (\Delta W_{t_j})^2 - 2t \sum_{t_k} (\Delta W_{t_k})^2 + t^2 \right] \\
 &\quad \begin{array}{c} \downarrow \text{4th moment of Gaussian: } 3\sigma^4 \\ \downarrow \quad \quad \quad \downarrow \quad \quad \downarrow \Sigma_{j \neq i} \end{array} \\
 &= \sum_i 3(\Delta t_i)^2 + \sum_i \Delta t_i (t - \Delta t_i) - 2t^2 + t^2 \\
 &= \sum_i 2(\Delta t_i)^2 \xrightarrow{\Delta t_i \rightarrow 0} 0
 \end{aligned}$$



Ito process

- Differential form $dX_t = b(X_t, t)dt + \sigma(X_t, t)dW_t$ ($X_t(\omega)$ is a continuous path for each sample ω)

- Integral form

$$X_t = X_0 + \sum_j \Delta X_{t_j} \quad t_j = j \cdot \Delta t \text{ and } \Delta t \rightarrow 0$$

$$= X_0 + \sum_j b(X_{t_j^*}, t_j^*) \Delta t + \sum_j \sigma(X_{t_j^*}, t_j^*) \Delta W_{t_j} \quad t_j^* \in [t_j, t_{j+1}]$$

Riemann integral

$$\int b(X_t(\omega), t) dt$$

?

Problem:

- $\int |dW_t| \approx \int \sqrt{dt} \rightarrow \infty$ (this term may not converge)
- Choice of t_j^* affects the result (W_t has high variation)

- Common choices of t_j^*

- Ito integral [left-end point]

$$t_j^* = t_j \Rightarrow \boxed{?} \text{ denoted by } \int \sigma(x, t) dW_t \text{ or } \int \sigma(t, \omega) dW_t(\omega)$$

Common form

Precise form indicating a r.v. for a sample ω

- Stratonovich integral [mid point]

$$t_j^* = (t_j + t_{j+1})/2 \Rightarrow \boxed{?} \text{ denoted by } \int \sigma(x, t) \circ dW_t \text{ or } \int \sigma(t, \omega) \circ dW_t(\omega)$$



About $\sigma(t, \omega)$

- Requirements

- Measurable: able to evaluate the probability of the image of $\sigma(t, \omega)$
- $\mathcal{F}_t$ -adaptable: $\sigma(t, \omega)$ depends only on the Brownian-motion sample ω , i.e. independent of future behaviors
- L2-Bounded:

$$\mathbf{E}_\omega \left[\int_s^T \sigma^2(t, \omega) dt \right] < \infty$$

- Properties

- Ito integral: $I[\sigma]$ is a r.v. and we choose the t-continuous version

$$I[\sigma](\omega) := \int_s^T \sigma(t, \omega) dW_t(\omega)$$

- Ito isometry

$$\mathbf{E}_\omega \left[(I[\sigma](\omega))^2 \right] = \mathbf{E}_\omega \left[\int_s^T \sigma^2(t, \omega) dt \right]$$

- Gaussian equivalence

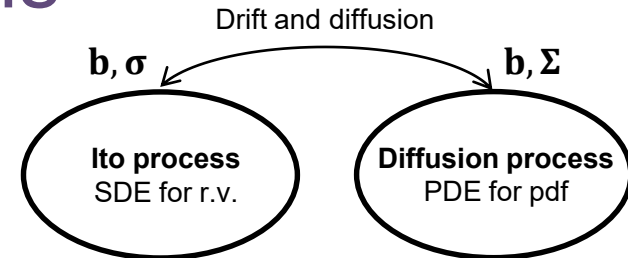
$$I[\sigma] \sim N \left(0, \mathbf{E}_\omega \left[\int_s^T \sigma^2(t, \omega) dt \right] \right) \quad (\text{almost surely})$$



Useful formulations

- Multi-dimensional extension

$$\underset{\mathbb{R}^n \uparrow}{d\mathbf{X}_t} = \underset{\mathbb{R}^n \uparrow}{\mathbf{b}(\mathbf{X}_t, t)}dt + \underset{\mathbb{R}^n \times \mathbb{R}^m \uparrow}{\boldsymbol{\sigma}(\mathbf{X}_t, t)}\underset{\mathbb{R}^m \uparrow}{d\mathbf{W}_t}$$



- Direct link to the equations of diffusion process, i.e. KFE, KBE, and RE, with drift $\mathbf{b}$ and diffusion $\boldsymbol{\Sigma} = \boldsymbol{\sigma}\boldsymbol{\sigma}^T$

- Ito formula

- Given a r.v. transformation $\underset{\uparrow \mathbb{R}^p}{\mathbf{Y}_t} = \mathbf{g}(\mathbf{X}_t, t)$, we have the following differential form for the k-th component of $\mathbf{Y}_t$

$$dY_{t,k} = \frac{\partial g_k}{\partial t} dt + \sum_i \frac{\partial g_k}{\partial X_{t,i}} dX_{t,i} + \underbrace{\frac{1}{2} \sum_{i,j} \frac{\partial^2 g_k}{\partial X_{t,i} \partial X_{t,j}} dX_{t,i} dX_{t,j}}_{\text{Special term due to } (dW_t)^2 = dt}$$

- Conversion of Stratonovich integral

- For the same $p(\mathbf{X}_t, t)$, Ito integral and Stratonovich integral can be considered as different interpretations of the drift term (important for physics)

$$d\mathbf{X}_t = \mathbf{b}(\mathbf{X}_t, t)dt + \boldsymbol{\sigma}(\mathbf{X}_t, t) \circ d\mathbf{W}_t \quad \equiv \quad d\mathbf{X}_t = \tilde{\mathbf{b}}(\mathbf{X}_t, t)dt + \boldsymbol{\sigma}(\mathbf{X}_t, t)d\mathbf{W}_t$$

If $\frac{\partial \sigma_{ij}}{\partial X_{t,k}} = 0$, the choice of t_j^* is irrelevant, and they are equivalent.

$$\tilde{\mathbf{b}} = \mathbf{b} + \frac{1}{2} \boldsymbol{\eta} \mathbf{1}, \eta_{i,j} = \sum_k \frac{\partial \sigma_{ij}}{\partial X_{t,k}} \sigma_{kj}$$



SDE for DM (1/2)

- Specific form of forward SDE: $d\mathbf{X}_t = f(t) \cdot \mathbf{X}_t dt + g(t)d\mathbf{W}_t$
 - Drift $\mathbf{b}(\mathbf{X}_t, t) = f(t) \cdot \mathbf{X}_t$ (Linear to $\mathbf{X}_t$)
 - Diffusion $\Sigma(\mathbf{X}_t, t) = g^2(t)\mathbf{I}$ (time-dependent, position-independent)

- Exact solution of probability density: $\mathbf{X}_t \sim N(\alpha(t)\mathbf{X}_0, \sigma^2(t)\mathbf{I})$
 - Direct conversion between $(f(t), g^2(t))$ and $(\alpha(t), \sigma^2(t))$
 - Proof with

$$\begin{cases} \text{Ito formula with } \mathbf{Y}_t = \mathbf{X}_t e^{-\int_0^t f(s)ds} \\ \text{Ito isometry for r.v. equivalence} \end{cases} \begin{cases} \alpha(t) := e^{\int_0^t f(s)ds} \Leftrightarrow f(t) = \frac{\alpha'(t)}{\alpha(t)} = \frac{d \log \alpha}{dt} \\ \sigma^2(t) := \alpha^2(t) \int_0^t \frac{g^2(s)}{\alpha^2(s)} ds \Leftrightarrow g^2(t) = \alpha^2(t) \frac{d}{dt} \left(\frac{\sigma^2(t)}{\alpha^2(t)} \right) \end{cases}$$

- ODE equivalence

KFE of forward SDE $\Rightarrow$

$$d\mathbf{X}_t = f(t) \cdot \mathbf{X}_t dt + g(t)d\mathbf{W}_t$$

KFE of ODE $\Leftarrow$

$$d\mathbf{X}_t = \mathbf{b}_{\text{ODE}}(\mathbf{X}_t, t)dt$$

$$\begin{aligned} \frac{\partial p(\mathbf{x}, t)}{\partial t} &= -\nabla_{\mathbf{x}}^T (f(t) \cdot \mathbf{x} \cdot p(\mathbf{x}, t)) + \frac{1}{2} g^2(t) \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p(\mathbf{x}, t) \\ &= -\nabla_{\mathbf{x}}^T \left(f(t) \cdot \mathbf{x} \cdot p(\mathbf{x}, t) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} p(\mathbf{x}, t) \right) \\ &= -\nabla_{\mathbf{x}}^T \left(\left(f(t) \cdot \mathbf{x} - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t) \right) \cdot p(\mathbf{x}, t) \right) \\ &= \mathbf{b}_{\text{ODE}}(\mathbf{x}, t) \end{aligned}$$



SDE for DM (2/2)

- Reverse SDE equivalence

KFE of forward SDE $\Rightarrow \frac{\partial p(\mathbf{x}, t)}{\partial t} = -\nabla_{\mathbf{x}}^T (f(t) \cdot \mathbf{x} \cdot p(\mathbf{x}, t)) + \frac{1}{2} g^2(t) \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p(\mathbf{x}, t)$

$$d\mathbf{X}_t = f(t) \cdot \mathbf{X}_t dt + g(t) d\mathbf{W}_t$$

$\times -1$

$$-\frac{\partial p(\mathbf{x}, t)}{\partial t} = \nabla_{\mathbf{x}}^T (f(t) \cdot \mathbf{x} \cdot p(\mathbf{x}, t)) - \frac{1}{2} g^2(t) \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p(\mathbf{x}, t)$$

$-g^2(t) \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p(\mathbf{x}, t)$

$+g^2(t) \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p(\mathbf{x}, t)$

RE of forward SDE $\Leftarrow = -\nabla_{\mathbf{x}}^T \left(\underbrace{-(f(t) \cdot \mathbf{x} - g^2(t) \nabla_{\mathbf{x}} \log p(\mathbf{x}, t))}_{= \bar{\mathbf{b}}(\mathbf{x}, t)} \right) \cdot p(\mathbf{x}, t) + \frac{1}{2} g^2(t) \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p(\mathbf{x}, t)$

Reverse time with $\tau = T - t$
and simplify symbols with $A(\cdot, t) = A^r(\cdot, \tau)$

KFE of reverse SDE $\Leftarrow$

$$\frac{\partial p^r}{\partial \tau} = -\nabla_{\mathbf{x}}^T (-\bar{\mathbf{b}}^r \cdot p^r) + \frac{1}{2} (g^r)^2 \nabla_{\mathbf{x}}^T \nabla_{\mathbf{x}} p^r$$

$$d\mathbf{X}_{\tau}^r = -\bar{\mathbf{b}}^r(\mathbf{X}_t, \tau) d\tau + g^r d\bar{\mathbf{W}}_{\tau}^r$$

Evolves with τ

$$d\bar{\mathbf{X}}_t = \bar{\mathbf{b}}(\mathbf{X}_t, t) dt + g(t) d\bar{\mathbf{W}}_t$$

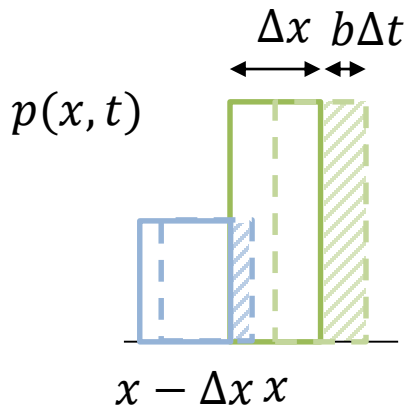
Equivalent form with t

Note: These SDE/ODE represent different stochastic processes but generate the same probability density. In practice, their results are different due to the errors brought by numerical integration and score estimation.



Intuitional reasoning

Drift ($dX_t = b(X_t, t) dt$):



$$dX_t = b(X_t, t) dt + g(t) dW_t$$

$\downarrow$ drift $\downarrow$ diffusion

$$\frac{\partial p}{\partial t} = - \frac{\partial}{\partial x} (b \cdot p) + \frac{1}{2} g^2(t) \frac{\partial^2 p}{\partial x^2}$$

$$p(x, t + \Delta t) \Delta x = p(x, t) \Delta x - p(x, t) \cdot b(x, t) \Delta t + p(x - \Delta x, t) \cdot b(x - \Delta x, t) \Delta t$$

$$\frac{\partial p}{\partial t} \Delta t \Delta x = - \frac{\partial}{\partial x} (b \cdot p) \Delta x \Delta t$$

Note: Pure drift is actually velocity field, and then this equation becomes Mass Conservation Formula.

Diffusion ($dX_t = g(t) dW_t$):

$$p(x, t + \Delta t) = p(x, t) \otimes g(t) \cdot n(x), n \sim N(0, \Delta t)$$

$$\downarrow \mathcal{F}_x$$

$$\hat{p}(\xi, t + \Delta t) = \hat{p}(\xi, t) \cdot e^{-2g^2 \Delta t \pi^2 \xi^2}$$

$$\downarrow \Delta t \rightarrow 0$$

$$\hat{p}(\xi, t + \Delta t) \approx \hat{p}(\xi, t) \cdot (1 - 2g^2 \Delta t \pi^2 \xi^2)$$

$$\downarrow \mathcal{F}_x^{-1}$$

$$p(x, t + \Delta t) \approx p(x, t) + \frac{1}{2} g^2(t) \Delta t \frac{\partial^2 p}{\partial x^2} \rightarrow \frac{\partial p}{\partial t} = \frac{1}{2} g^2(t) \frac{\partial^2 p}{\partial x^2}$$

$$\frac{d^2}{dx^2} f(x) \xleftrightarrow{\mathcal{F}_x} -4\pi^2 \xi^2 \hat{f}$$

On reverse generation paths

- A sample of reverse process corresponds to its initial x_T and its path
 - SDE generation
 - The path is related to per-step Gaussian randomness
 - Each x_T has an infinite number of paths (one x_T maps to x_0 probabilistically)
 - ODE generation:
 - The path is deterministic
 - Each x_T corresponds to one path exactly (one x_T maps to one x_0)

